VISHAKAN BIOTECH
PROSPERITY THROUGH
BIOLOGICALS

VISHAKAN BIOTECH



PRODUCT BROCHURE

INDIAN MADE GLOBALLY ADMIRED

PROSPERITY THROUGH BIOLOGICALS



- Bio Fertilizers
- Bio Insecticides
- Growth Promoters

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Other Products

1000+ Customers | 50+ Products | 5 Awards won

About us

Vishakan Biotech Pvt Ltd (VBT) is a professionally managed company specialized in Biological solutions for Agriculture and environmental applications. VBT has the widest range over 50 plus products for pest and disease management, plant health & nutrient and soil health management. Our products are renewable and residue free for high quality and efficacy. VBT focused on research and technology, we have R & D laboratory with necessary infrastructure, Agricultural scientists, Microbiologists, Pathologists and Agronomist. Our research team are dedicated to quality and innovation. Our products will help the customers around the world to control diseases, pests, improve plant and soil health & increase crop yields. It helps to reducing the chemical pesticide residues in food and other agricultural produce.

Our Competence

- Isolation and identifying the most effective microbial strains and enhancing its efficacy in many folds.
- Most strain are isolated from the concerned area soil, (adapted to the local environment/soil)
- Scientific Production method with high cfu, zero contamination and long shelf life.
- Producing various formulations (liquid, powder, granules)
- Isolates are highly tolerable in soil pH, Temperature, Humidity etc, and high survival rate in the field.
- Our products are organic certified by INDOCERT and biofertilizer under Fertilizers Control Order (FCO).

Our Mission

- Offering sustained quality of our products and unsurpassed in service excellence
- Developing and maintaining our high level of integrity, long term sustainable and transporting business relationship
- The maintenance of a reputation as an honest dependable supplier as our range of products
- Working closely with our supplier and clients in the pro-active development of new products and ideas.

Our Team

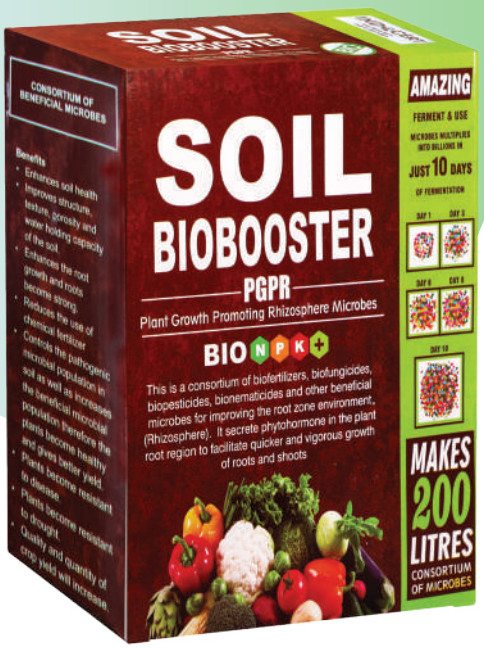
Dr.P.Thamizhchelvan who has done his Doctorate in Microbiology and has a wealth of experience in Agricultural Microbiology for more than 29 years. He served as R&D head and Director-Technical in agriculture bio-products manufacturing companies. He developed and commercialized various proprietary microbial formulations for soil and plant health management, pest and disease control, yield enhancement etc., One of his formulation is a microbial consortium is very successful in commercial and domestic Solid Waste Management. Now he is the Managing Director of Vishakan Biotech Private Limited.

Dr.D.Purusothaman is former Professor head, Department of Agricultural Microbiology TNAU, Coimbatore is the technical advisor to lead our company in various scientific activity and innovative programme.

Dr.L.Manivel former TRA, Tocklai & UPASI Plant Physiologist is our Scientific Technical Advisor.
He has wide experience and knowledge on Horticulture and Plantations crops.

Lets
Grow with
Bio Farming

PROPRIETARY PRODUCTS



SOIL BIOBOOSTER

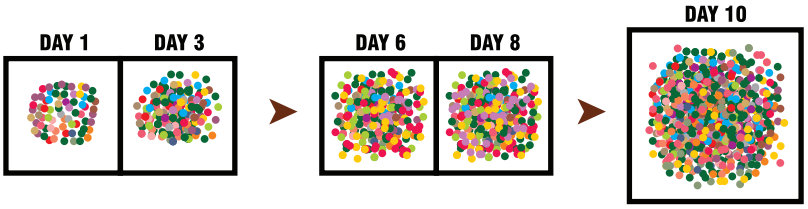
Soil Biobooster is a combination of Biofertlizer,Biopesticide, Bionematicide and few beneficial microbes. It help to induce plant growth, controls various plant disease, pests, insects and nematodes etc.

The beneficial microbes secrete phytohormones in the plant root region to facilitate quicker and vigorous growth of root and shoot.

Crops

Horticulture Crops, Bedding Crops, Container Crops, Agriculture Crops, Flowering Crops, Ornamental Crops Truffgrass, Trees, Spices

Ferment & Use Microbes Multiples Into Billions In Just **10 DAYS** Of Fermentation



Preparation

- Mix 1Kg Microbial consortium and 1½ Kg Starter in 200 Lts of water in a plastic container
- Allow to ferment for 10 days
- Stir well at leastonce a day
- Spray / drench in the morning or evening hours
- For best results avoid chemical fertilizers, pesticides or fungicides before or after 10 days of application

MAKES 200 LITRES CONSORTIUM OF MICROBES

Dosage

After fermentation, drench 250ml to 5 ltrs (depends upon the size and age of the plant) around the root zone of the plant followed by light irrigation. For best result apply in the early morning or evening.

Benefits

- Enhances soil health
- Improves structure, texture, porosity and water holding capacity of the soil
- Enhances the root growth and roots become strong
  The plants become resistant to disease
- Reduces the use of chemical fertilizer
- Controls the pathogenic microbial population in soil as well as increases the beneficial microbial population therefore the plants become healthy and gives better yield
- The plants become resistant to drought

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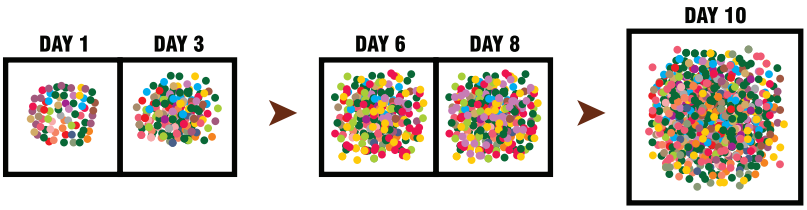
ROT ARREST

BIO CONTROL FOR FUNGAL DISEASES

Crops

Root Rot, Rhizome Rot, Red Rot,
Leaf Blight, Leaf Fungal, Foliar Diseases

Ferment & Use 10 DAYS
Microbes Multiples
Into Billions In Just
Of Fermentation



How to Use

Mix 1Kg microbes and 1/2 Kg Starter in 200 Lts of water in a plastic container, allow to ferment for 10 days, Stir well at least one time a day.

MAKES 200 LITRES
CONSORTIUM OF MICROBES

Soil Application

2 Pkts/Acre (400 Lt water), 1 Lit V- KITIN. After 10 days of fermentation mix 1 Lit V-KITIN and drench / drip around the root zone of the plant followed by light irrigation. If the infection is more higher dosage may be recommended

Foliar Application

1 Pkt/Acre (200 Lit water). After 10 days of fermentation spray in the foliar region throughly

Time

Apply in the early morning or evening hours.

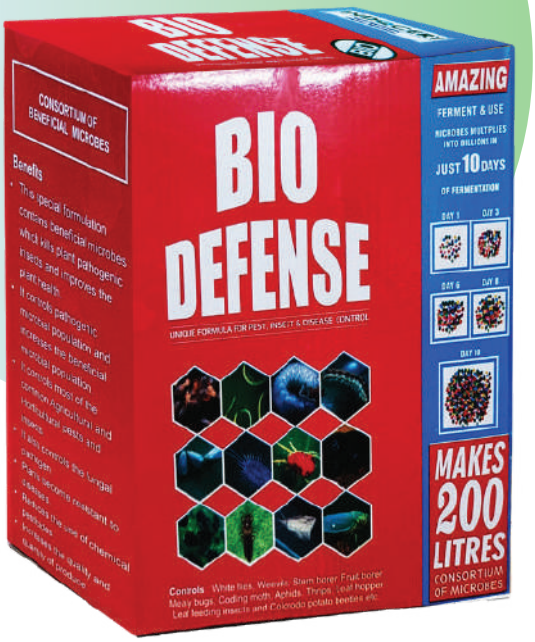
Duration

30-45 days Interval

Caution

Do not mix with chemical fungicides

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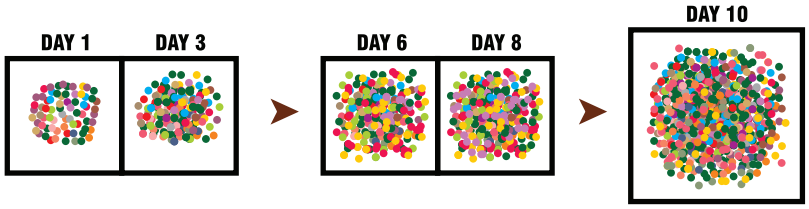
BIO DEFENSE

Bio Defense is a combination of Bio-pesticide, Bio-fungicide and few beneficial microbes. It helps to induce plant growth, Controls various plant diseases, Pests and Insects.The beneficial microbes secrete phytohormones. It facilitates quicker and vigorous growth of plants.

Crops

Horticulture Crops, Agriculture Crops, Truffgrass, Bedding Plants, Flowering Crops, Trees, Container Plants, Ornamental Crops, Spices

Ferment & Use Microbes Multiples Into Billions In Just Of Fermentation **10 DAYS**



Benefits

- It controls most of the common Agricultural and Horticultural pests and insects
- This special formulation contains spores of the beneficial microbes, which kills the plant pathogenic pests and insects and induces plant growth.
- The plants become resistant to diseases
- Reduces the use of chemical Pesticide
- Controls the pathogenic microbial population as well as increases the beneficial microbial population.
- Controls house fly & mosquitos in domestic environment.
- Increases the quality and quantity of produce

MAKES 200 LITRES CONSORTIUM OF MICROBES

Target pests

White flies, weevils, stem borer, fruit borer, mealy bugs, codling moth, aphids, Thrips, Leaf hopper, Leaf feeding insects and Colorado potato beetles etc.

Dosage

After fermentation, drench or spray 250ml to 5 ltrs (depends upon the size and age of the plant) around the root zone of the plant & leaves followed by light irrigation. For best result apply in the early morning or evening.

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YIELDER PLUS

Yielderplus is a brassinolide based growth stimulant. It contains Bio Active Macro & Micronutrients,Cytokinines, Auxin, Gibberlines, Brassinolides, Flower inducing Hormones and other Biostimulants in liquid form.This product accelerate cell division and cell elongation. Besides this, it is synergetic effect with many plant growth regulators. It is very effective in low concentration.

Yielderplus greatly enhance the plants ability to uptake nutrients giving them the boost they need to launch them into growth. One to three more treatments throughout the crop cycle can greatly enhance the plants ability to reach its optimum potential.

Yielderplus is safe and sustainable

Crops

Flowering Crops, Bedding Plants, Agriculture Crops, Truffgrass, Horticulture Crops, Container Plants, Spices, Ornamental Cropss, Treea

Dosage & Application

500ml/Acre or 3 to 5ml/Lit Water

Benefits

- It increases the photosynthesis activity.
- It increases the size and weight of the fruits.
- More and bigger blossoms
- Greater root and leaf growth
- Improves soil conditions
- It promotes early and rapid germination of seeds.
- It can be mixed with chemical fertilizers, pesticides and fungicides.

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V-ZOOT

Natural pest & insect control



V - Zoot is an botanical based bio pesticide which is medium to broad spectrum with different mode of action.

The primary role is disrupting an insect's metamorphosis influences the hormonal system, feeding activity, reproduction and flying ability of insects. It disrupts the different stages of life cycle of insects.

V-Zoot has strong odour which acts as an insect repellant. It is effectively prevents insects from sucking or eating plants.

It has low side effects on non-target species and beneficial insects.

Multiple modes of action make insects and plant pathogens cannot develop resistance.

Benefits

- It controls almost all pests & insects like whitefly aphids, mealybugs, thrips, caterpillars, leafhopper, mites, scales etc.,
- It is eco-friendly and residue free
- This can also be used in IPM program

Dosage & Usage

2-3ml / Lit Water. Apply preferably in the early morning or late evening hours

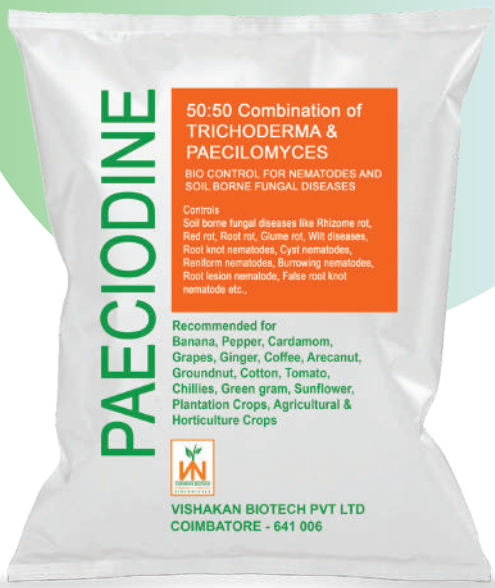
Compatibility

V-ZOOT is compatible with most of the agriculture crop protection chemicals and Bio-control agents.

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PAECIODINE

Paeciodine controls root knot nematodes, cyst nematodes, reniform nematodes, borrowing nematodes, root lesion nematodes, false root nematodes. Also controls secondary infection of soil borne fungi diseases like rhizome rot, red rot, root rot, clump rot, wilt diseases.



Crops

Banana, Pepper, Cardamom, Grapes, Ginger, Coffee, Arecanut,Groundnut, Cotton, Tomato, Chillies, Green gram, Sunfower Plantation Crops, Agriculture & Horticulture Crops

Method of Application

- Nursery: Mix 1 Kg of PAECIODINE in 200 Ltr water & drench 25 cents nursery area.
- Seed Treatment: For 1 Kg of seeds, treat with 10 gm of PAECIODINE before sowing.
- Plant Pit Application: Before transplanting the seedlings, apply 50 gms of PAECIODINE around the root zone of plant before monsoon.

Main Field Application

1 Kg of PAECIODINE mixed in 200 Ltr water, mix well and drench
2 to 5 Ltr per plant depend upon the age and size of the plant.

BIO-FERTILIZERS

V-NITRO

Azospirillum
Nitrogen Fixing Biofertlizer



V-NITRO efficient strain of Azospirillium Azospirillium is a nitrogen fixing free-living bacterium. It is found in the soil around the plant roots and root surface. Nitrogen is the major element present in the atmosphere. But it is present in an unavailable form to the plants. This has to be made available to the plants for their basic physiological activities.

Mode of action

- V-NITRO assimilates atmospheric nitrogen (which can not be taken by plants directly) and then fixes it in soil, which is then readily available to the plant.
- V-NITRO is known to produce Plant growth promoting substances like IAA, GA and Cytokinins, which promotes root proliferation, plant growth, early initiation of tillers retention of more number of flowers and capsules.
- V-NITRO produces vitamins like Pantothenic acid, Thiamin and Niacin in large quantities.
- These substances improve the plant growth and yield.
- V-NITRO inoculation promotes early tillering and reproductive growth and increases the grain-filling rate.
- V-NITRO can fix 30-50 Kg of Nitrogen per hectare in one-crop season and can leave behind substantial quantities of nitrogen in the soil for succeeding crop.
- V-NITRO increases the rootlet density and root branching resulting in the increased uptake of minerals and water.

Crops

Rice, Sugarcane, Millets, Coconut, Cotton, Banana, Cardamom,
Coffee, Rubber, Tea, Vanilla and all Agricultural and Horticultural crops.

Method of application

Seed treatment

For Paddy (1 Kg/acre or 250ml/acre) other crops (1/2 Kg/acre or 125ml/acre). Mix required quantity of V-NITRO with 250ml of cool rice kanji and make a slurry. Treat the seeds sufficient for an acre. Dry in shade for 30 minutes and sow.

Seedling treatment

Mix 2 Kg or 500ml of V-NITRO in 50 lit of water and dip the roots for 15 mts before transplanting.

Main field application

Mix 5 Kg or 1000ml of V-NITRO in 100 Kg of well composted farmyard manure and broadcast in one acre of main field, just before transplanting.

V-NITRO should not be mixed with any chemical insecticide, fungicide and weedicide. Maintain organic matter and adequate moisture before and after application

Viable cell count : 5 x 10^7 / gm (Powder)
1 x 10^8 / ml (Liquid)
Best before 6 months from manufacturing date

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V-PHOS

Bacillus megaterium
Phosphate Solubilizing Bacteria



Phosphorus is one of the essential nutrient required for plant growth.

V-PHOS contains microbial strains of phosphate solubilizing bacteria (Bacillus megaterium), which solubulize insoluble phosphates in soils and makes it available to plant.

Mode of action

- V-PHOS, which dissolve, insoluble form of phosphorus to solubulize and available to plants is called phosphobacteria
- V-PHOS also produces amino acids, Vitamins and growth promoting substances like IAA, GA, which help in better growth of the plant
- Phosphate is required by the plant for the activities of food synthesis, Flower formation, Fruit setting and seed setting. Phosphorus nutrition is absolutely essential for cell reproduction and growth of root and shoot.
- Inoculation of V-PHOS into soil produces organic acids such as Formic, Acetic, propionic, Lactic, Glycolic, Fumaric and Succinic acids. Some of these acids form chelates with cations such as Ca and Fe and such chelation results in effective solubilization of phosphates.
- V-PHOS increases 17-20% of the crop yield.

Crops

Rice, Ragi, Maize, Sorghum, Arecanut, Sugarcane, Millets, Coconut, Cotton, Cocoa, Banana, Cardamom, Pepper, Coffee, Rubber, Tea, Vanilla, Orange, Amla, Turmeric and all Agricultural and Horticultural crops.

Method of application

For Paddy (1 Kg/acre or 250ml/acre) other crops (½ Kg/acre or 125ml/acre). Mix required quantity of V-PHOS with 250ml of cool rice kanji and make a slurry. Treat the seeds sufficient for an acre. Dry in shade for 30 minutes and sow.

Seedling Treatment

Mix 2 Kg or 500ml of V-PHOS in 50 lit of water and dip the roots for 15 mins before transplanting.

Main field application

Mix 5 Kg or 1000ml of V-PHOS in 100 Kg of well composted farmyard manure and broadcast in one acre of main field, just before transplanting.

V-PHOS should not be mixed with any chemical insecticide, fungicide and weedicide. Maintain organic mater and adequate moisture before and after application.

Viable cell count : 5 x 10^7 / gm (Powder)
1 x 10^8 / ml (Liquid)
Best before 6 months from manufacturing date

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V-SOL-K

Frateuria aurantia
Potash Mobilizing Bacteria



Potassium is one of the essential nutrients required for plant growth.

V-SOL-K contains efficient strain of Frateuria aurantia, which dissolve insoluble form of potash in soil to make it available to plant.

Mode of action

- Most of the potassium in soil exists in the form of silicate minerals.
- Potash solubulizing bacteria produces bacterial acids, alkalies, chelants and exopolysaccharides which helps release of potassium from silicate.
- Potash is required for the activities of food synthesis, Flower formation, Fruit setting and seed setting.
- V-SOL-K encourages early root development which hastens the maturity and there by increases the Yield
- V-SOL-K increases 17-20% of the crop yield

Crops

Rice, Ragi, Maize, Sorghum, Areca nut, Sugarcane, Millets, Coconut, Cotton, Cocoa, Banana, Cardamom, Pepper, Ginger, Coffee, Rubber, Tea, Vanilla, Orange, Amla, Turmeric and all Agricultural and Horticultural crops.

Method of application

Seed Treatment

For Paddy (1 Kg/acre or 250ml/acre) other crops (1½ Kg/acre or 125ml/acre). Mix required quantity of V-SOL-K with 250ml of cool rice kanji and make a slurry. Treat the seeds sufficient for an acre. Dry in shade for 30 minutes and sow.

Seedling Treatment

Mix 2 Kg or 500ml of V-SOL-K in 50 lit of water and dip the roots for 15 mins before transplanting.

Main field application

Mix 5 Kg or 1000ml of V-SOL-K in 100 Kg of well composted farmyard manure and broadcast in one acre of main field, just before transplanting.

V-SOL-K should not be mixed with any chemical insecticide, fungicide and weedicide. Maintain organic mater and adequate moisture before and after application.

Viable cell count : 5 x 10^7 / gm (Powder)
1 x 10^8 / ml (Liquid)
Best before 6 months from manufacturing date

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V-ZINC

Bacillus sp
Zinc Solubilizing Bacteria



Zinc is one of the essential nutrients required for plant growth. V-ZINC contains efficient strain of Zinc solublizing bacteria, which dissolve insoluble form of Zinc in soil to make it available to plant

Importance of Zinc

- In production of auxin, an essential plant growth hormone
  Regulates starch formation and proper root development
  In formation of chlorophyll and carbohydrates
- Enable plants to withstand lower air temperature
- Helps in the biosynthesis of cytochrome; a pigment, and maintains plasma membrane integrity, and helps synthesis of leaf cuticle

Crops

Rice, Ragi, Maize, Sorghum, Areca nut, Sugarcane, Millets, Coconut, Cotton, Cocoa, Banana, Cardamom, Pepper, Ginger, Coffee, Rubber, Tea, Vanilla, Orange, Amla, Turmeric and all Agricultural and Horticultural crops.

Mode of action

Zinc solubilizing bacterium multiplies, secrete organic acids and helps to solubilize insoluble and chelated zinc fixed in the soil. This mobilized zinc is easily available to the plants

Method of application

Seed Treatment

For Paddy 1 Kg/acre,500ml/acre other crops ½ Kg/acre or 250ml/acre .Mix required quantity of inoculent with 250ml of cool rice kanji and make a slurry. Treat the seeds sufficient for an acre. Dry in shade for 30 minutes and sow.

Seedling treatment

Mix 2 Kg or 500ml of V-ZINC in 50 lit of water and dip the roots for 15 mins before transplanting

Main field application

Mix 5 Kg or 1000ml of V-ZINC in 100 Kg of well composted farmyard manure and broadcast in one acre of main field, just before transplanting.

V-ZINC should not be mixed with any chemical insecticide, fungicide and weedicide.
Maintain organic matter and adequate moisture before and after application.

Viable cell count : 5 x 10^7 / gm (Powder)
1 x 10^8 / ml (Liquid)
Best before 6 months from manufacturing date

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V-RIZHO

Rhizobium Nitrogen Fixing Bio Fertilizer



V-RHIZO is a biological N supplimenter contains efficient strain of Rhizobium. Rhizobium is symbiotic nitrogen fixing bacterium living in association with leguminous crops and assimilates atmospheri nitrogen and fixes in the root nodules

Role of V-Rhizo

- V-Rhizo normally enters into root hairs of plant, multiply there and produces nodules and fixes the atmospheric nitrogen.
- Leguminous plants take the nitrogen fixed by V-Rhizo
- The root secretions from the leguminous plants and biological material released from the decomposed nodule, enriches the fertility status of soil.
- It can fix 50-100 Kg of nitrogen per hectare in one crop season and can leave behind substantial quantities of nitrogen for succeeding crop.
- It can increase yield from 15-20%.

Crops

Black gram, Green gram, Red gram, Bengal gram, Soyabean, Groundnut and other leguminous crops.

Method of application

Seed Treatment

Mix required quantity of (1/2 Kg/Acre or 125ml/Acre)
V-RHIZO with cool rice kanji and make a slurry.
Treat the seeds sufficient for an acre. Dry in shade for 30 minutes and sow.

Main field application

5Kg or 1000ml of V-RHIZO in 100 Kg of well composted farmyard manure and broadcast in one acre of main field, just before transplanting.

Points to be noted

V-RHIZO should not be mix with any chemical insecticide, fungicide and weedicide.
Maintain adequate moisture and organic matter before and after application.

Viable cell count : 5 x 10^7 / gm (Powder)
1 x 10^8 / ml (Liquid)
Best before 6 months from manufacturing date

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V-RICH

Vesicular Arbuscular Mychorrhizae



V-RICH is VAM fungi (Vesicular Arbuscular Mychorrhizae). The symbiotic association between plant roots and fungal mycelia is termed as mycorrhizae. It acts as an extension of root absorbing surface. These fungi can improve the plant growth through increased nutrient uptake of phosphorus, sulphur, calcium and zinc.

VAM - genera

Glomus sp., Gigaspora sp., Acaulospora sp, Scutellospora sp. & Sclerocystis sp.

Mode of action - uptake of nutrients

The P-ions are readily adsorbed on clay complexes and diffuse slowly in the soil so that a depletion zone quickly develops around the actively absorbing rootlets. Fungal hyphae can explore the soil beyond the depletion zone and transport phosphate back into the root The phosphate is translocated into the mycelium in the root and is released for use by plants. Further these fungi are known to enhance resistance to diseases and help the host plant to take up moisture more efficiently under moisture stress condition. VAM fungi transports N, K, Ca, Mg along the fungal mycelium in a manner similar to phosphorus.

Crops

Pepper, Cardamom, Ragi, Citrus, Clove, Tea, Coffee, Cocoa, Banana, Pepper, Rubber, Mango, Apple and other perennial crops

Benefits of V-RICH

- Increases nutrient and water uptake
- Improves soil and plant ecosystem
- Increases plant growth and establishment
- Increases yield
- Reduces transplant stress and plant loss
- Improves soil structure and porosity
- Reduces the need for chemical fertilizers
- Improves drought tolerance
- Improves resistance to soil borne pathogens like Phythium sp, Fusarium Sp, Rhizoctonia sp, Phytopthroa etc.

Method of application

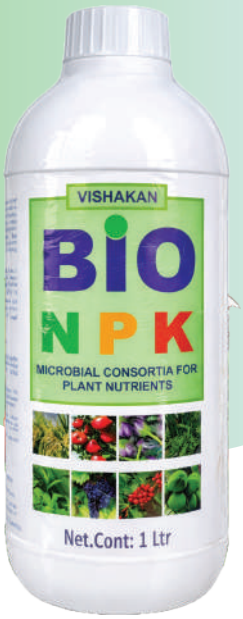
Nursery: In nurseries this should be applied 2-3 cm below the top soil at the time of sowing.
Existing plant: For existing crops 20-200 gm inoculam is required depending on plant size.
Before application crack the topsoil and maintain enough moisture and organic matter.

Viable propagules: 10 /g
Infective Potential: 1200 IP/gm
Best before 6 months from manufacturing date

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BIO-NPK

Microbial consortia for plant nutrients



VISHAKAN BIO-NPK (liquid) is microbial formulation containing strains of bacteria able to synthesize/assimilate atmospheric nitrogen, solublize phosphate and potash into available form thereby supplementing balance nutrition to the crops. It converts non-available forms of macro & micro nutrients in to available form.

Application:

Drip Irrigation

500ml - 1.0Lit per Acre (1.25 Lt to 2.5 Lt per ha)

Seed treatment

5-10ml/Kg of seed - Mix in water 1:19 ratio. Soak the seeds for 30 minutes, dry in shade, sow after 2-3 hours in the evening.

Seedling root dipping

Mix 250-500ml of Premium Bio-NPK in 20-30 Litres of water and dip the roots for 30 minutes before transplanting then transplant seedling.

Mainfield application

Mix 1.0 Lt/Acre of Vishakan Bio-NPK in sufficient quantity of soil or orgranic manure/compost and broadcast in the field before last ploughing.

In Standing Crop

Broadcast the above mixture and irrigate the field immediately.

Recommended Crops

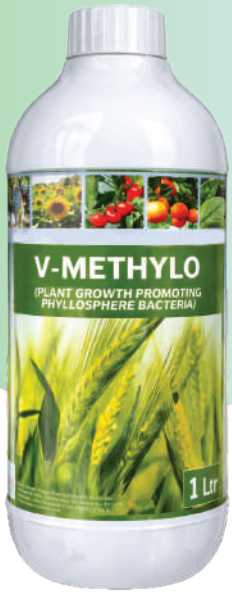
All agricultural and horticultural crops

Advantages

- Improves plant pH
- Make available essential elements to plant
- Good germination, better roots & plant growth
- Improves quality and yield of the crop

V-METHYLO

Phyllosphere bacteria



V-METHYLO contains exclusive strain of methylobacterium sp., characterized by high drought resistance, cytokinin production, stimulate seed germination, produces phytohormones, improves the chlorophyll intensity and increase leaf area thus improves the yield.

Application

Seed Treatment:

10ml/Kg seeds. Soak the seeds for 30 mins at 1% V-Methylo solution and sow the seeds.

Seedling root dipping:

Mix 10ml of V-Methylo in 1 Lit of water and dip the roots for atleast 10 minutes before planting.

Foliar Application:

500ml-1000ml/Acre. Spray at morning & Evening hours.

Soil Application:

1000ml/Acre of V-Methylo mixed with requied amount of Farm Manure and applied.

Benefits

- **Fastens seed germination & seedling growth.**
- **Helps the crop to withstand drought for 25-30 days and for crop recovery after rain (or) irrigation.**
- **Earliness in flowering, fruit setting and maturation.**
- **Improves fruit colour, quality, size and weight.**
- **Mitigate drought & increases the yield.**

POPULATION LOAD: $1*10^{9}$ cfu/ml.

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BIO FUNGICIDE

VIRIDINE

Trichoderma Viride
Biological Fungicide



VIRIDINE is a biological fungicide. Which contains efficient strains of Trichoderma viride. A naturally occurring mycopathogenic (fungus antagonist) fungus that infects and limits the growth of plant pathogenic fungi. These formulations are specifically manufactured to enhance the mycocidal activity and biological stability of the conidia.

The formulation helps to control the rot, Rhizome rot, foot rot and collar rot, wilt and damping off diseases caused by phytopathogenic fungi to crop plants.

Mode of action

- It produces certain plant growth promoting substances which help better plant growth.
- It forms a natural protective barrier around the root zone against pathogen entry

Benifits of VIRIDINE

It is an effective bio-control agent against seed and soil borne diseases.
It is effective against Rhizoctonia solani, Phythium Sp, Macrophomina phaseolina, Sclerotina Sp, etc.

Crops

Sugarcane, Pulses, Apple, Oilseeds, Cotton, Vegetables, Banana, Coconut, Oil palm, Chillies, Lime, Coffee & Tea, Areca nut & Rubber, Flower crops, Spices, & Condiments etc., all Vegetable crops and Horticultural crops

Method of application

Seed Treatment

6g or 3ml per Kg of seeds as dry seed treatment and broadcast in main field.

Seedling root dipping

(1Kg or 250ml per acre)

Soil Treatment

5Kg or 1000ml of VIRIDINE to be mixed with 100 Kgs farmyard manure and apply in one acre of main field

Compatibility

VIRIDINE is compatible with most bio fungicides

Points to be noted

VIRIDINE should not be mixed with any chemical insecticide, fungicide and weedicide.

Viable cell count: 2x 10^6/gm (Carrier)
Viable cell count: 2x10^6/ml (Liquid)
Best before 6 months from manufacturing date

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V-CURE

Pseudomonass fluoresens



V-CURE is a biological fungicide. The technical name is Pseudomonas fluoresens, a naturally occurring mycopathogenic bacterial antagonist that infects and limits the growth of plant pathogenic fungi.

These formulations are specifically manufactured to enhance the mycocidal activity and biological stability.

The formulation helps to control leaf, bud, rhizome and root rot, wilt, Blast, Sheath blight, Stem bleeding, Soya bean Pustule, Citrus canker etc.

These organisms are not harmful to the crop but inhibit the growth of other disease causing fungi.

Mode of action

- Colonizing the root system of the host plant
- Producing secondary metabolites namely siderophores, which can suppress the growth of parasitic pathogen by restricting the iron availability to the pathogen.
- Its ability to compete for nutrients and space with other microorganisms is another important characteristic of Pseudomonas, which makes it a good alternative bio-control agent.
- It has ability to produce plant growth promoting substances, which promote the plant growth and improved nutrient uptake.
- Enhanced stress resistance in plants and helps increase the quality and quantity of yield.

Crops

Banana, Cardamom, Pepper, Ginger, Rubber, Vanilla, Coconut, Groundnut, Paddy, Apple, Pulses, Turmeric, Tomato and other vegetable crops etc.

Method of application

Seed Treatment

10g or 5ml per Kg of seeds and soak the seeds in water over night, decant the excess water and allow to sprout the seeds for 24 hrs and then sow.

Seedling root dipping

2.5 Kg or 500ml of V-CURE for 1 ha seedlings in sufficient amount of water then dip the roots for at least 30 minutes.

Foliar Application

Spray 0.4% concentration (4g or 2ml/lit 2 Kg or 500ml/ha) commencing from 45 days after transplanting at 10 days interval for 3 times depending on disease intensity. If there is no disease incidence a single spray is sufficient as a preventive.

Points to be noted

Before application crack the topsoil and apply around root zone and maintain moisture.
V-CURE should not be mixed with any chemical insecticide, fungicide and weedicide.
Maintain adequate organic matter and moisture before and after application. .

Viable cell count: $1x10^8$ /gm (Carrier)
Viable cell count: $1x10^8$ /ml (Liquid)
Best before 6 months from manufacturing date

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V-BACILI

Bacillus subtilis
Biological fungicide



V-BACILI is a biological bactericide. The technical name is Bacillus subtilis, a naturally occurring mycopathogenic bacterial antagonist that infects and limits the growth of plant pathogenic fungi & produces very powerful fungal cell wall busting super bio-enzymes.

These formulations are specifically manufactured to enhance the mycocidal activity and biological stability.

The formulation helps to control and protects the crops from both baterial and fungal diseases in grapes, pomegranate, banana, cardamom, paddy, pulses, vegetables and other agriculture & horticulture crops. This is very effective for powdery mildew.
These organisms are not harmful to the crop but inhibit the growth of other disease causing fungi.

Mode of action

- **Colonizing the root system of the host plant**
- **It produce very powerful fungal cell wall busting super bio enzyme.**
- **It has ability to produce plant growth promoting substances, which promote the plant growth and improved nutrient uptake.**
- **Enhanced stress resistance in plants and helps increase the quality and quantity of yield.**

Crops

Banana, Cardamom, Grapes, Pomegranate, Pepper, Ginger, Rubber, Apple, Vanilla, Coconut, Groundnut, Paddy, Pulses, Turmeric, Tomato and other vegetable crops etc.

Method of application

Seed Treatment

10g or 5ml per Kg of seeds and soak the seeds in water over night, decant the excess water and allow to sprout the seeds for 24 hrs and then sow.

Seedling root dipping

2 5 Kg or 500ml of V-BACILI for 1 ha seedlings in sufficient amount of water then dip the roots for at least 30 minutes

Foliar Application

Spray 0.4% concentration (4g or 2ml/lit, 2 Kg or 500ml/ha) commencing from 45 days after transplanting at 10 days interval for 3 times depending on disease intensity. If there is no disease incidence a single spray is sufficient as a preventive.

Points to be noted

Before application crack the topsoil and apply around root zone and maintain moisture.
V-BACILI should not be mixed with any chemical insecticide, fungicide and weedicide.
Maintain adequate organic matter and moisture before and after application.

Viable cell count: 2 x 10^8/gm (Carrier)
Viable cell count: 1 x10^8/ml (Liquid)
Best before 6 months from manufacturing date

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BIO PESTICIDE

V-CIDE

Beauveria bassiana



V-CIDE is a biological insecticide. The technical name is Beauveria bassiana. Beauveria bassiana is a naturally occurring entamopathogenic fungi, which causes disease known as the white muscadine disease in insects. The fungus does not infect humans and other animals and is considered as an insecticide. These formulations are specifically manufactured to enhance the insecticidal activity and biological stability.

Mode of action - uptake of nutrients

When the spores of the fungus come in contact with the cuticle (skin) of the susceptible insects, they germinate and grow directly through the cuticle into the inner body of the host. The fungus proliferates throughout the body, producing toxins and draining the insect of nutrients eventually killing it.

Target pests

Stem borer, weevils, aphids, Thrips, Termites, Leaf hopper, Leaf feeding insects, Colorado potato beetles, corn borer, White flies, and codling moth, mealy bugs and Mites etc.

Crops

Most of the crops including cotton, Paddy, Tomato, Tea, Coffee, Cardamom, Pepper, Ginger, Apple, Soybeans, Green house crops and other Horticultural and Agricultural crops.

Method of application: Foliar spray 5 gm/Lit or 2 ml/Lit

Mix the required amount of V-CIDE along with required amount of water , stir well to form an uniform suspension. Spray immediately after preparation. Spraying should be done during late afternoon or early morning hours. Thorough coverage of both sides of leaf surfaces is must to get excellent results. For speedy action, higher dosage may be applied. Use natural wetting agent for better results.

Soil Application

Dissolve (2 Kg or 1 Lit V-CIDE in 400 lit water and drench around the root zone area.

Compatibility

V-CIDE is compatible with most bio-pesticides and conventional insecticides. It should not be mixed with any chemical fungicides.

Storage

Keep in dry and cool place. Do not expose to direct sunlight or heat.

Viable cell count: 1 x10^8/gm (Carrier)
Viable cell count: 1 x10^8/ml (Liquid)
Best before 6 months from manufacturing date

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V-FIGHT

Metarrhizium anisopliae



V-FIGHT is a biological insecticide. The technical name is Metarrhizium anisopliae. Metarrhizium is a naturally occurring entamopathogenic fungi, which causes the disease known as the white muscadine disease in insects. The fungus does not infect humans and other animals and is considered as an insecticide.
These formulations are specifically manufactured to enhance the insecticidal activity and biological stability.

Mode of action - uptake of nutrients

When the spores of the fungus come in contact with the cuticle (skin) of the susceptible insects, they germinate and grow directly through the cuticle into the inner body of the host. The fungus proliferates throughout the body, producing toxins and draining the insect of nutrients eventually killing it. Many insect pathogens release metabolites that increases the chance of death of insect. Destruxens from Metarrhizium are polypeptides that appear to be directly toxic to the insect.

Target pests

Root grubs, Termites, white flies, weevils, aphids, Thrips, Leaf hopper, Leaf feeding insects, Colorado potato beetles, Corn borer, Stem borer, and Codling moth, Mealy bugs etc.

Crops

Most of the crops including cotton, Paddy, Tomato, Tea, Coffee, Cardamom, Pepper, Apple, Ginger, Soybeans, Green house crops and other Horticultural and Agricultural crops.

Method of application : Foliar spray 5 gm/Lit or 2 ml/Lit

Mix the required amount of V-FIGHT along with required amount of water , stir well to form a uniform suspension. Spray immediately after preparation. Spraying should be done during late afternoon or early morning hours. Thorough coverage of both sides of leaf surfaces is a must to get excellent results. For speedy action, higher dosage may be applied. Use natural wetting agent for better results.

Soil Application

Dissolve 2 Kg or 1 Lit of V-FIGHT in 400 lit water and apply around the root zone area.

Compatibility

V-FIGHT is compatible with most bio-pesticides and conventional insecticides.
It should not be mixed with any chemical fungicides.

Storage

Keep in dry and cool place. Do not expose to direct sunlight or heat.

Viable cell count: 1 x10^8/gm (Carrier)
Viable cell count: 1 x10^8/ml (Liquid)
Best before 6 months from manufacturing date

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V-KILL

Verticillium lecanii
Biological Insecticide



- V-KILL is a biological insecticide. The technical name is Verticillium lecanii.
- Verticillium is a naturally occurring entamopathogenic fungi, which causes disease known as the white muscadine disease in insects.
- The fungus does not infect humans and other animals and is considered as an insecticide.
- These formulations are specifically manufactured to enhance the insecticidal activity and biological stability.

Mode of action

When the spores of the fungus come in contact with the cuticle (skin) of the susceptible insects, they germinate and grow directly through the cuticle into the inner body of the host. The fungus proliferates throughout the body, producing toxins and draining the insect of nutrients eventually killing it.

Target pests

White flies, weevils, aphids, Thrips, Red spider mite, Potato beetles, Corn borer, Stem borer, and Codling moth, Mealy bugs.

Crops

Most of the crops including cotton, Paddy, Tomato, Tea, Coffee, Cardamom, Pepper, Ginger, Soybeans, Green house crops and other Horticultural and Agricultural crops.

Method of application: Foliar spray 5 gm/Lit or 2 ml/Lit

Mix the required amount of V-KILL along with required amount of water , stir well to form an uniform suspension. Spray immediately after preparation. Spraying should be done during late afternoon or early morning hours. Thorough coverage of both sides of leaf surfaces is must to get excellent results. For speedy action, higher dosage may be applied. Use natural wetting agent for better results.

Compatibility

V-KILL is compatible with most bio-pesticides and conventional insecticides. It should not be mixed with any chemical fungicides

Dosage

1Kg or 500ml V-KILL in 200 liters of water for one acre spray.

Storage

Keep in dry and cool place. Do not expose to direct sunlight or heat.

Viable cell count: 1 x10^8/gm (Carrier)
Viable cell count: 1 x10^8/ml (Liquid)
Best before 6 months from manufacturing date

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BIO NEMATICIDE

V-NEMATO

Paecilomyces lilancinus
Biological Nematicide



V-NEMATO is a biological Nematicide. The technical name is Paecilomyces lilancinus Paecilomyces has the ability to control number of nematodes such as Root Knot nematode. Burrowing nematode, Cyst nematode, Root lesion nematode, Reniform nematode, False root knot nematode.

Mode of action

During the infection of nematodes, the nematophagous fungi penetrate the surface (Cuticle) of the nematode and digest the internal tissues of the host .The nematode cuticle is composed of proteins including collagen. Proteolytic enzymes produced from the fungus have an important role during the penetration. These fungi parasitize vermiform adult or larval nematodes and parasitize eggs or cyst.

Crops

Vegetables, Banana, Pineapples, Grapes, Strawberries, Citrus, Peach, Ornamentals, Tobacco, Nut trees and other Agricultural and Horticultural crops

Dosage

Dissolve 1 Kg or 500ml of V-NEMATO in 400 lit water drench around base of each plant/around the root zone area.

Method of application

Mix the required amount of V-NEMATO along with required amount of water, stir well to form a uniform suspension. Spray / drench around base of the each plant immediately after preparation.

Spraying should be done during early morning or evening hours. For speedy kill, higher dosage may be applied. Use natural wetting agent for better results.

Timing

In general 14 days before planting, 6 weeks after planting, repeat every 6 weeks for 4 months

Compatibility

V-NEMATO is compatible with most bio-pesticide and bio-fertilizers. It should not be mixed with any chemical fungicides and pesticides.

Storage

Keep in dry and cool place. Do not expose to direct sunlight or heat.

Viable cell count: 2 x 10^7/gm (Carrier)
Viable cell count: 2 x10^8/ml (Liquid)
Best before 6 months from manufacturing date

BIOSTIMULANTS

V-COMBINE

Bio stimulant for plants



V-Combine is natural bio stimulant for plants, root and soil

Advantages of V-Combine

- V-Combine can be foliar and soil application
- Balanced nutrient composition of V-Combine in natural form promotes general health of crops and enhances defense mechanisms of plants against invading diseases pests and natural imbalances e.g.,drought and frost
- During the flowering stage application of V-Combine enlarges fruit size, increases the yield and improves the quality of produce.
- A chelating agent and soil enhancer.
- Increases resistance against crop diseases

Direction for use: Soil application: Apply directly to the soil with the aid of a conventional garden sprayer (Dilute 1 lit of V-Combine in 500 lit water /acre spray)

Soil drenching: 4 ml/lit water and drench around the root zone area.

Foliar application: 2 ml/lit water. Spray should cover both surfaces of leaves and flowers

Crops	Time Of Application
Fruits Flowers, Orchards Citrus, lemon, orange, Grape, Guava, Amla Papaya, Mango, Chiku, Apple, Roses etc.	Pre flowering stage Just after flowering At fruiting period
Vegetables Tomato, Potato, Chillies, Cuccumber, Onion, Cabbage, Beans, etc	Time of sowing Before transplanting 10 days before flowering At frutting period (2-3 sprays at 10 days interval
Cereals Maize Paddy, Wheat, Bajara etc	Time of sowing Before transplanting Growing stage panicle initiation stage 20 days after I spray
Pulses Soybean, Green gram etc	Before sowing Growing and Flowering stage Before flowering 20 days after flowering
Oil seeds Ground nut, Mustard, Sunflower etc	Time of sowing Growing and Flowering stage Before flowering 20 days after flowering
Plantation crops Tea, Rubber, Turmeric, Ginger,coffee, Tobacco	Each spray with a monthly interval preferably during high crop period
Sugarcane	Time of sowing Time of sugar formation stage

Shelf Life: 3 years from manufacturing date, generally stable for very long time

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V-COMBINE GRANULES



V-COMBINE-GR is innovative biotechnology research. A unique formulation product by mixing humic, amino, seaweed, vitamins and other natural plant hormones in scientifically controlled conditions to give you the soil and plant health and optimum output profit.

V-COMBINE-GR is a natural balanced granulated version of the nutrient composition

Benefits

- It promotes general health of crops
- It enhances defense mechanisms of plants against invading diseases, pests and natural imbalances e.g., drought and frost.
- It enlarges fruit size, increases yield and improves the quality of produce.

V-COMBINE-GR is chelating agent and soil enhancer

Dosage

Main field Application - 10kg/Acre
Nursery Application - 2KG/ Acre

Composition

Seaweed (Ascophyllum nodosum) extract per cent, by weight, minimum	-	0.83
Protein hydrolysates derived from gelatin through chemical Hydrolysis with calcium hydroxide and neutralized with sulphuric acid per cent by weight, minimum	-	0.33
Humic acid (Source: Leonardite) per cent. by weight, minimum	-	1.3
Calcium hydrogen phosphate (as inert) per cent. by weight, minimum	-	1.0
Dolomite granules (as carrier) per cent. by weight, minimum	-	96.54
Total (per cent.)	-	100

Specification

Alginic acid per cent, by weight, minimum	-	0.08
Total amino acid per cent, minimum	-	012
Humic acid per cent.by weight, minimum	-	0.57
pH(1% solution)	-	7.78 + 0.5
Total organic carbon per cent. by weight, minimum	-	6.11
Total dissolved solids per cent. by weight, minimum	-	0.79
Bulk density (g/cc)	-	1.1491 + 0.02
Solubility per cent. by weight,minimum	-	30.26

V-HUME



V-Hume is a dark brown /black colour liquid derived from decayed organic matter (humates).

Active ingredient

Humic acid and fulvic acid 12.5%, other plant growth promoter's and major and minor nutrients.

Mode of action

V-Hume has high cation exchange capacity which serves to chelate (collect) plant food elements and release them as the plant requires.

Recommendation

Seed Treatment: Mix 10 ml of V-Hume per Kg of seeds with required quantity of water
Seedlings & Root tip: Mix 2 ml V-Hume/litre and dip the roots for 5 mts before planting

Foliar spraying: Apply 1 ml/litre of water and spray during flower and fruit formation stages.

Soil drenching: Mix 3 ml V-Hume /litre of water and drench nursery bed, Root zone area
Drip & Sprinkle irrigation: 1 lit for one acre

Crops

V-Hume is recommended for all categories of crops like Cereals, Pulses, oil seeds, Vegetables, fruit crops, plantation crops and all horticulture crops.

Composition

Humic acid per cent. by weight minimum	-	10.0
Fulvic acid per cent. by weight minimum	-	2.5
pH(10% aqueous solution)	-	12-14
Specific Gravity	-	1.0 - 1.2

Benefits

- Vigorous seed germination
- Effectively chelates and complexes soil micro nutrients.
- Increases soil aeration, water holding capacity and stimulates growth and proliferation of desirable soil microorganisms.
- Enhances nutrient uptake increases yield and quality of produce Breaks up unproductive clay soils and turns into profitable soils, Increase soil microorganism's metabolic activity
- Stimulates shoot and root growth by accelerating cell division
- Regulates crop growth and stimulate enzyme and plant growth hormone synthesis like Auxins, Cytokinines.
- Increases vitamin content in plants, carotinoid content in carrots and grain weight in rice, wheat, pulses etc.
- Reduces soil erosion and improves drought tolerance.
- V-Hume is suitable for all soil and crops
- Stimulates the respiration rates, Increase root and top growth on a fresh & dry weight basis & enhance plant root uptake of P, K, Fe, Cu, Zn and Ca.

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V-HUME GRANULES

Soil & plant enhancer



V-Hume-GR is a slow granulated version of the humic materials.

V-Hume-GR increases soil fertility, humic content and metabolic activity of soil microbes. Increases nutrient absorption and chelates metals effectively. Shoot and root growth is much better. Increases quality of the produce and provides better stress tolerance to plants.

V-Hume-GR increases number of tillers, panicles, grains per panicles, and grain weight.

Crops

V-Hume-GR is very effective for crops like Banana, Paddy, Sugarcanes, Wheat, Cotton, Chilles, Cardamom, Pepper, Coffee, Rubber, Tea, Vanilla, Cocoa, Pomegranate, Mango, Apple, Oranges, Grapes, Sapota and all other agricultural and horticultural crops.

Dosage

Main field application -10Kg/Acr
Nursery application - 2Kg/Acre

Composition

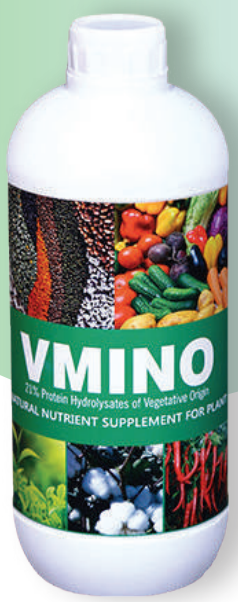
Humic Substance (Source: Renewable agri-biomass) percent by weight minimum	-	1.5
Bentonite per cent.by weight, minimum	-	98.5
Total (per cent)	-	100

Specification

Humic acid per cent, by weight, minimum	-	0.05
Fulvic acid per cent, by weight, minimum	-	0.03
pH (1% solution)	-	7.0-9.0
Bulk density (g/ml)	-	0.90-1.10
Total organic carbon per cent, by weight, minimum	-	0.436

V-MINO

Plant extract for flower & fruit setting



V-mino is naturally hydrolysed protein of vegetables origin (20%) which provide excellent growth for all crops like Cereals, Pulses, Oil seeds, Vegetables, Fruit crops, Plantation crops.

Vmino's role and Benefits in Plants

- Increases supply of macro and micro nutrients
- Provides enzymatic activities and natural stimulants
- Multiplies soil micro flora and fauna due to its role as bio-activator.
- Stimulates photosynthesis due to increased availability of CO_2
- Stimulates protein synthesis as it contains amino acids which act as building blocks.
- Increases resistance against crop diseases
- Increases flowering, fruit setting and stops pre mature flower dropping
- Provides drought resistance by its direct activity.
- Improves quality and quantity of produce

Recommendation

Foliar spray: 500-1000 ml/acre spray. Spray should cover both surfaces of leaves and flowers.:

Composition

Ingredient		Content (%)
Protein hydrolysate derived from soybean through acid hydrolysis (minimum, % by weight)	-	20
Diethylene Glycol (minimum, % by weight)	-	8
Xanthum gum (minimum, % by weight)	-	8
Surfactant (Envipol 3010) (minimum, % by weight)	-	5
Citric acid (minimum, % by weight)	-	2
Water (% by weight)	-	QS
Total	-	100

V-ZYME

Ascophyllum nodosum 15% Liquid

V-Zyme is a liquid sea weed extract. It contains Biologically hydrolysed protein complexes, Plant growth promoters.



Importance: V-Zyme provides wide variety of chemical, physical and biological improvements to the crops and its growing environment. It contains important nutrients, eg: Nitrogen, Phosphorous, Potassium, Zinc, Iron, Magnesium, Manganese, Boron, Copper and growth promoters etc. v-zyme promotes general health of crops and enhances defence mechanisms of plants against invading diseases, pests and natural imbalances.

Crops

V-Zyme is recommended for all categories of crops like Rice, Sugarcane, Greengram Millets, Soyabean, Cotton, Banana, Cardamom, Coffee, Rubber, Tea, Vanilla, Nutmeg, Cocoa, Pineapple and all Agricultural and Horticultural crops.

Recommendation

500-1000ml/acre Foliar spray or Root Zone application

Benefits

- It provides N,P,K as well as trace minerals e.g.Zn, Mn, Mg, Fe, etc and growth promoters in natural, balanced and chelated form.
- Because of balanced nutrient composition it promotes general health of plant including drought and frost resistance.
- Promotes stronger stem and leaf growth. It accelerates photosynthesis and further develops healthy foliage.
- Promotes faster growth of the plant.
- It shows enhanced resistance to some fungi,molds, aphids,mites and scabs that attack plant at variou stage.
- Induces flowering, Prevent shedding of plants and fall of unripe fruits.
- Enlarged fruit size, increases the yield and improves the quality of produce.
- V-Zyme contains many sea life metabolites, algal sugars, iodine,etc at proper ratio.
- It can inhibit powdery mildew and other diseases.

Composition

Alginic acid per cent, by weight minimum	-	1.5
Mannitol per cent. by weight minimum	-	1.0
pH(10% aqueous solution)	-	4.7 ± 1.0
Specific Gravity	-	1.0 - 1.1
Total organic carbon per cent, by weight, minimum	-	5

Shelf life: Effective life is 3 years from date of manufacturing under normal storage conditions.

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V-ZYME GR

Seaweed Extract Granules



ZYME-GR is based on seaweed (Sargassum tenerrimum 2%) and combination of marine algaes and finest marine plant available for agriculture use and is recounted worldwide as an excllent natural fertilizer and source of organic matter V-ZYME-GR application enables plants to receive direct benefits from the naturally balanced nutrients and plant growth substances available in the seaweed extract

Features

- ZYME-GR provides over 60 naturally occurring major and minor nutrients and plant development substances comprising of enzymes, proteins. cytokinins, amino acids, vitamins, gibberellins. Auxis, betains etc. in organic form.
- V-ZYME-GR provides all constituents in balanced form for healthier plant growth.
- V-ZYME-GR contributes to greater microbial activity when applied to soil and thus increasing the nutrient availability to plants.
- V-ZYME-GR is an ideal organic product for better growth and productivity, which can be used on all types of plants, whether indoor, outdoor, garden, nursery, lawns, turf, agriculture or plantation crops.

Application Method

Field Crops- Broadcast V-ZYME-GR granules uniformly on thick sown crops. For wide spacing crops, use furrows/spots/fertigation methods of application.

Vegetables, Fruits, Plantation crops- Apply V-ZYME-GR in furrows or in spots around the plants, mix with the soil and then water the plants/field.

Flowers & pot plants- Apply 8-10 grams of V-ZYME-GR around each plant, 2 to 3 inches away from the stem. After application, earth up the soil to mix the granuels with top layer and water the plants normally.

Turf and Lawns- Broad cast 40 to 50 grams V-ZYME-GR per square meter before planting or 20 to 25 grams after planting and water the lawn Repeat the granular application once in every six month

Composition

Alginic acid per cent, by weight minimum	-	0.2
pH (1:2 aqueous solution)	-	6.0 - 8.5
Bulk density (g/cc)	-	0.95 - 1.1

Direction of use

Try to use entire quantity of V-ZYME-GR once the pack is opened. Store at room temperature and avoid prolonged exposure to direct sun Timing and rate of application is very important for maximum efficacy hence crop-wise schedule and guidelines should be followed carefully.

Avoid application of herbicides in combinatie with V-ZYME-GR

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V-ZYME GEL

Seaweed Extract Gel



Y-ZYME GEL is 100% water soluble pure seaweed pulp. This seaweed pulp is a natural combination of micro nutrients, macro nutrients, secondary nutrients and plant growth harmones that helps to increase the intake of dissolved nutrients,which reduces the fertilizer runoff, V-ZYME GEL increase the chlorophyil formation, photosynthesis process, it helps to prevent the nutrients heat stress and deficiency.

V-ZYME GEL is a soil conditioner that enriches your agricultural land and makes the soil fertile, thus, improving the immune system of the plants. Hence, V-ZYME GEL increases the quality and the quantity of the yields products from your land.

Our V-ZYME GEL is the only product with all 3 plant growth hormones, that helps the plants in flower formation, root formation and stem formation. It contains Seaweed extract, macro, micro & secondary nutrients, alginic, proteins hydrolyzed, vitamins & minerals

Features

- **V-ZYME GEL can be converted for liquid formulation and Bentonite Coating (1:4 Ratio - Kg diluted with 4 ltrs of water)**
- **It increases the soil water holding capacity upto 30%, Resist against unforseen condition.**
- **It helps plants in the intake of atmospheric nitrogen that is essential in the healthy growth of crops.**
- **It aids in reducing the amount of usage of synthetic fertilizer**
- **Our seaweed gel makes soil more fertile when used in less / unfertile soil.**
- **Enlarged root system, enlarged leaf structure and excellent seed germination.**
- **Crop yield and quality is increased upto 30%**
- **Enriches the soil and making the root colour into its natural white.**

Dosage

Drip: for short term crop 2-3Kg per acre
Basal application: for short term crop - 2-3Kg per acre
Drip: for long term crop 5-6Kg per acre
Basal application: for long term crop - 5-6Kg per acre

Appearence

Brownish Black

Odour

Fishy

Storage

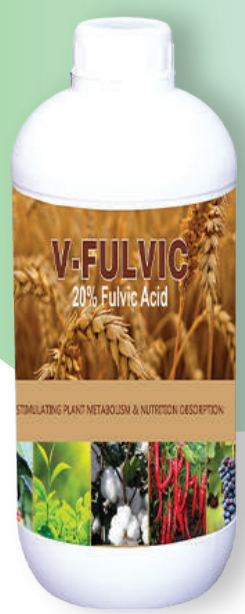
Keep it in cool & dry place
Keep away from children & pets

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V-FULVIC

Stimulating Plant Metabolism & Nutrition Absorption

Fulvic substance extracts from natural sources



Application:

Seed Treatment

10ml fulvic per kg of seed with required quantity of water

Seedling & Root dip

2ml per lit of water

Spraying

1ml per ltr water
Drip & Sprinkler Irrigation: 1 Ltr/acre



Benefits

- Vigorous seed germination & root growth
- Enhances nutrient absorption
- Increases microbial activity
- Efficient utilization of nutrients
- Stimulates enzyme formation
- Regulates crop growth
- Increases quality and quantity of the produce
- Better stress tolerance

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DECOMPOSERS

V-COMPOST

Consortium of composting microbes



V-COMPOST is a microbial consortium which is used to compost domestic and agriculture waste.

The microbes are isolated from Indian soil which contains bacteria's, actinomyces etc., This microbes are aerobic in nature and are not genetically modified organisms.

Preparing V-COMPOST for composting
For 1 ton agriculture waste, mix 2 Kgs of V-COMPOST + 2Kgs of jaggery + 25 to 50 Kgs fresh cow dung. Mix 50-75 Ltr water. This solution is used for composting process.

How to use

Direct Application in Plants.

Chose a well elevated and shaded area and arrange the compostable residues in layer by layer to form a heap. (i.e., compostable residue as 1st layer and sprinkle v-compost solution over the 1st layer, then spread the 2nd layer of compostable residue over the 1st layer and sprinkle v-compost solution over the 2nd layer). Maximum layers is 4

The height of the layer should not exceed 1 feet and the total height of the heap should not exceed 4 to 5 feet.

During the composting process adequate moisture should be maintained, when ever needed the heap should be hydrated. While hydrating water should not leach from the heap. During rainy seasons the heap should be protected from rain

Normally the processing time is 45 to 60 days. It depends up on the nature of the materials, woody and hard materials will take longer time to compost.

Benefits

- **Increases the soil texture and fertility**
- **It help to increase the available nutrient level to the plants**
- **Increase the soil moisture and aeration of the soil.**
- **It increases the organic carbon level of the soil**
- **Reduces the usage of chemical fertilizers**
- **Reduces the environmental problems**
- **Since the organic matter in the compost can hold more water so the ground water storage increases**
- **Increases the beneficial microbial population in the soil**
- **Reduces the harmful microbial population of the soil so the pest and disease infection will reduce**
- **It helps to increase the NPK availability to the plants**
- **Reduce the quantum of the domestic and agriculture waste**

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BMC

Beneficial Microbial Consortium



V-BMC is a unique combination of different types of bacteria, yeast and fungi. It has been thoroughly tested and proven safe for humans and animals. V-BMC suppresses pathogens in soil and accelerates organic decomposition.

It increases the availability of mineral and useful organic compounds to plants, and helps to increase the proliferation of beneficial microorganismsin the soil.

Preparation Method

1 Litre of solution is prepared by mixing 10ml of BMC, 40ml of molasses or 120 gram of Jaggery and 950ml of water and leaving it for 5 to 7 days depending on the temperature. This liquid is then mixed with 1 litre of molasses or 3kg of Jaggery and 98 litres of water. Now 100 litres of V-BMC liquid is ready for use.

Method Of Use

Foliar Application

Leaf Blight can be avoided by using a clean sprayer weekly in the morning or afternoon.

Soil Application

By spraying the solution on mature fallen plants and dried plant waste in the open ground and keeping the soil moist, organic matter increases in the soil.

Application Interval

3 to 4 months interval

Recommended

All types of domestic & agricultural wastes

V-CLEAN

Aerobic composting media maker



V-Clean: This special formulation contains strains of various composting microbes derived from the natural sources and combined together to form a consortiumin a unique method.

Starter is a unique blend of food for the composting microbes. It is specifically designed for the particular fermentation and to be used with the V-Clean composting microorganisms only.

During fermentation V-Clean microbes secretes various enzymes, which aids the composting process and also multiplies into billions.

After fermentation this liquid canbe infused into processed or unprocessed coirpith (low EC or high EC)

After infusion of V-Clean microbes, normal coirpith becomes an efficient Aerobic Composting Media. This media can be used in any type of aerobic composting method/devices for making compost.

Cost Effective

Each 5Kg coirpith block will yield approx. 60Ltrs of hydrated aerobic composting media
Each 650gm coirpith brick will yield approx. 9 Ltrs of hydrated aerobic composting media.
This pack is sufficient for infusing 7Nos of 5Kg blocks, which will yield approx 400-450 Ltrs of aerobic composting media.
(or)
This pack is sufficient for infusing 50 Nos of 650gm bricks, which will yield approx 400-450 Ltrs of aerobic composting media.

MICRONUTRIENTS

V-NUTRINOL

Micronutrient Mixture VI for Vegetables



Plants required Micro-Nutrients for satisfactory metabolic production to exhibit their maximum potential apart from preventing from pests and diseases.

Benefits

- Induces more flowering
- Reduces flower pod and fruit shedding
- Reduces deficiency diseases
- Promote healthy plants
- Enhance yield

Recommendation

It is recommended for citrus, Cardamom, Guava, Sapota, Tea, Coffee, Papaya, Mango, Grapes, Cashew and vegetables.

A wholesome product for successful entrepreneur.

Composition

Nutrients		%
Iron	(Fe)	7.60
Manganese	(Mn)	1.22
Zinc	(Zn)	1.68
Boron	(B)	2.48
Molybdenum	(Mo)	0.14
Copper	(Cu)	1.00



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COCONUTRI

Micronutrient Mixture IV for Vegetables

Multi micro nutrient fertilizer provides essential nutrients to balance these deficiencies



Benefits

- Improved tree health and growth.
- Increased coconut yield.
- Enhanced resistance to pests and diseases.

Application

500 grams to 1000 grams per treeper year.

Application Intervals

3 to 6 months interval

Composition

Nutrients		%
Iron	(Fe)	3.80
Manganese	(Mn)	4.80
Zinc	(Zn)	5.00
Boron	(B)	1.60
Copper	(Cu)	0.50



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OTHER PRODUCTS

V-SILICA



V-SILICA is a liquid potassium silicate fertilizer designed to increase crop pest resistance reduce evaporation losses from plants and produce

Dosage of Foliar Application Vegetables

1ml/Lit water in 15 days interval. 2 to 3 spray require

Plantation Crops

- 1 to 1.5ml/Lit of water in 30 days interval.
- 3 to 4 spray required in one year.

Orchards & Vine Yards

- 1 to 1.25ml/Lit water in 45 days interval.
- 2 to 3 spray required.

Soil Application

1ml/Lit water and apply as required.

Benefits

- Increase photosynthesis and raise brix in all plants.
- It reduces the unwanted excess aluminum, sodium and manganese.
- Improves plant growth and increase the quality and quantity of the yield.
- Enhances the reproduction by improving pollenation and increasing pollen fertility.
- Produces thickening the outer cell wall resulting in stronger, more resilient plants.
- Ideal for use on crops to prevent to reduce water stress (drought) symptoms.
- Reduces insect damage to crops due to higher plant strength.

Store in a cool dry area out of direct sunlight. Keep out of reach of children

V-CHILL

Plant Stress Stablizer



An anti-stress crop solution based on fatty acid alcohol that cools foliage by reducing plant tissue temperature and reflecting harmful sun radiation.

V-Chill works predominantly by scattering sun rays. It generates a thin glassy filmcoat that reflects incident light more than would take place in normal conditions, when applied as foliar spray. This stops light from having thermic action on plant tissues.

Direction For Use

Prepare the spray solution by placing the specified amount of V-Chill into a dry, clean container, then add water and stir thoroughly to achieve a proper emulsion.

Benefits

- Minimize water loss from leaf surface.
- Increase resistance to drought and frost.
- Increase quality & durability of the product.
- Enchance rejuvenation from thermal or cold frost.
- Helps to maintain moisture content of plant cell.
- Provide better yield even under moisture stress condition.

Dosage

3ml/Lit (Depends upon crop canopy)

Composition	m/m
Fatty Alchocol and Diluent	80%
Emulsifier	5%
Neutraliser	10%
Stabliser	5%
Total	100%

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V-KITIN



V-KITIN is a unique formulation based on Natural Marine Chitosan Oligisaccarides.

It work as a natural plant defence booster that builds up plant immunity to fight against pathogens and support the plant to overcome environmental stress, therefore speeds up germination and growth, improves the quality and quantity of the yield.

V-Mode Of Action

- Chitosan is a linear polysaccharide composed of randomly distributed B-(1-4)-linked D-Glucosamine and N-acetyl-D-Glucosamine.
- V-KITIN induces the synthesis of Pytoalexins and several other enzymes for the plant defence mechanism and also for the recovery of the plant after pathogenic attack.

Active Ingradient

Chitosan Oligosaccarides

Direction To Use

- Foliar Application: 2-2.5 ml/Lit of Water
- Drenching: 1 ml/Lit of Water and drench around root zone area.

Compatibility

Compatible with Bio-Fertilzers, Organic fertilizers, Organic manure, Bio-Pesticides, Vermi-compost, Seaweed fertilizer & Protien hydrolysates.

Crops

For all type of Agricultural, horticultural, flowering crops, Cereals, Pulses & Oilseeds.

Benefits

- Stimulates immunity in crops & degrade pesticide residue.
- Helps the plant to develop new roots and shoots.
- It also helps in stem elongation and blooming.
- Induces the chlorophyll content in the plant which leads to high degree of photosynthesis
- Helps in vigorous germination of seeds.

Shelf life

3 Years from manufacturing date generally stable for very long time.
Shake well before use. Store at room temperature

V-BIOPOTASH

Fruit & Blossoms enhancer



V-BIOPOTASH is Potash based organic Fertilizer which is specifically designed to overcome potassium deficiency in plants. This helps to increase flower formation, fruit & seed set, Fruit colour and rate of maturity resulting in high yield.

V-Mode Of Action:

Potassium is an essential nutrients required in large amount by plants. Potassium regulates water balance, nutrients in plant tissues, increases protein synthesis and disease resistance.

V-BIOPOTASH contains active ingredients that chelates with the minerals and activates plant enzymes which helps in the synthesis of Carbohydrates, amino acids & proteins hence results in better absorption of nutrients by plants for a longer duration.

Composition

Potash-14%
Protein hydrolysate -30%
Organic carbon->50%

Direction To Use

- **Foliar Application: 2-3 ml/Lit of Water**
- **Drip irrigation: 1-1.5ml/Lit of water**

Crops

Paddy, Wheat, Maize, Fruits, Vegetables, Cotton, Cereals and all other agricultural and horticultural crops.

Benefits

- **Strengthens plant cell wall, hence develop disease & Pest resistance in plants.**
- **increase photosynthesis & water use efficiency in plants.**
- **improves soil structure, porosity and fertility of the soil.**
- **Improves the shelf life of the yield.**

Shelf life

3 Years from manufacturing date generally stable for very long time.
Shake well before use. Store at room temperature

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WET SPREAD



Wet Spread is a Silicon based spray adjuvant. It gives super spreading & dramatically reduces surface tension of spray solution.

Therefore the performance of the pesticides & fertilizers is significantly enhanced by a) Superior wetting of hydrophobic surfaces, particularly leaves & other plant parts. b) Better absorption of nutrients & systemic pesticides through cuticular waxes. c) Stomatal flooding.

Strictly follow the label instructions on Weedicides, Insecticides, Fungicides & Plant growth regulators for tank filling. Make only 90% of the desired volume by adding relevant agrochemical.

Take 10% of the remaining water. Add required quantity of Wet Spread. Stir throughly, wait & add this to main tank to make up the volume to 100%.

Since weather, crops, soil and other conditions may vary, neither we nor the seller make warranty of any kind, expressed or implied, concerning the use of this products. The user assumes all risk of use and handling whether or not in accordance with directions or suggestions.

Composition

Applcation with	Dosage
Contact action pesticide	0.15ml/lit
Systemic action pesticide	0.25ml/lit
Fertlizers & Plant Growth Regulators	0.15ml/lit

V-NIMBIDINE



V-NIMBIDINE is a neem oil based biopesticide containing Azadirachtin 0.03% EC w/w min. Neem oil based antifeedant/pest repellent preparation has been prepared to control aphids and bollworm of cotton and stem borer and leaf folder of rice.

Features

- **V-NIMBIDINE contains azadirachtin 300ppm, which controls a wide range of pests including whiteflies, aphids, thrips, caterpillars, leafhoppers, etc.**
- **It effectively controls target pests.**
- **Safe for beneficial insects such as parasites, predators, and honeybees.**

Mode of action

Nimbecidine 300ppm acts as an antifeedant, repellent, ovi-position deterrent, insect growth regulator, and sterilant.

Recommendation:

Common name of pest	Dosage	Dilution in water litre	Waiting period last spray to harvest day
Aphids	2.5	500	5
Helicoverpa armigera (Bollworm)	2.5	500	5
Leaf Folder	2	500	5
Stem borer	2	500	5

V-MIRACLE



V-MIRACLE is a powerful solution designed to protect the crop from destructive insects like Aphids, White Flies, thrips, hoppers etc..
V-Miracle spreads and acts fast resulting in quick control of insects. through permanant feeding inhibition.It affects all stages of pests life cycle from the immature stage to adult stage.

Benefits

- V-MIRACLE is having good phytotonic property and promote the growth and yield of crop
- Low mammalian toxicity combined with highly selective target pest.
- It is effective in minimum dosage level, cost effective and eco-friendly.
- Excellent sunlight stability, rain resistant properties, providing extended residual activity.
- It has powerful and long-lasting effect and has excellent translaminar action to control pests in the lower leaf surface.

Dosage

0.5 ml / Lit of water

Composition	Percent
Herbal Concentrate	10%
Preservative & Stablizer	10%
Filler & Others	80%

V-ATTACK



V-ATTACK is a unique blend of herbal plant extracts that are known to have insecticidal properties. It is formulated in such a way that the pesticide adhere to the pests and penetrate their exoskeletons. It is effective against a wide range of pests including Larvae, Fruit Borer, Aphids, Whiteflies, Jassids, Thrips and Mealy Bugs.

Benefits

- It have good phytotonic property and promote the growth and yield of the crops
- It boosts the immunity of the plant to fight against pests.
- Attack is very effective of minimum dosage level.
- It is cost effective, eco-friendly, non-toxic.
- It is effective on the pests that are hidden inside plants.

Dosage

0.5ml / Lit of water

Composition	Percent
Herbal Extract	10%
Stablizer	10%
Aqua box with emulsifier	80%

V-KRISHONATE



Shake well before use, Dilute 2-3 ml V-Krishiyan in one liter of water
Contents: Potassium Phosphonate

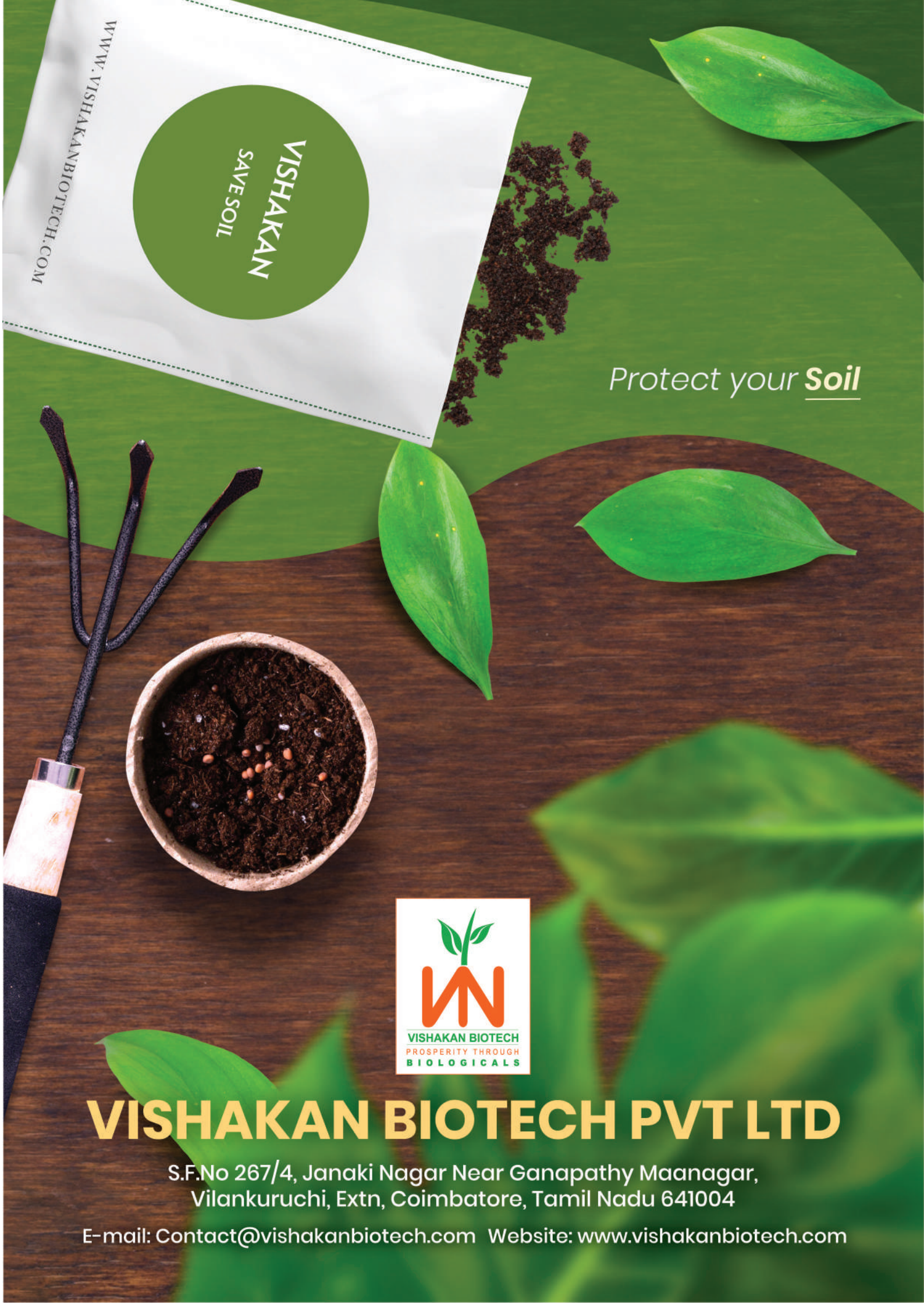
Recommendations:

Crops	Dosage	Time of application
Cardamom	3ml/Liter 3ml/Liter	Pre and post monsoon For spray - High humidity with low temperature weather condition For drenching
Ginger & Turmeric	5ml/Liter	
Black Pepper	3ml/Liter	Pre and post monsoon For drenching
Arecanut	5ml/Liter	Pre Monsoon

General Information:

V-Krishiyan is water soluble and non-toxic to plants when applied in accordance with the directions





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